Materia:
Integración de seguridad informática en redes y sistemas de software (Gpo 601)

Actividad evaluativa:
7.4.2 Laboratorio - Implementar DHCPv4

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Dispositivo	Interfaz	Dirección IP	Máscara de subred	Gateway predeterminado
R1	G0/0/0	10.0.0.1	255.255.255.252	N/D
	G0/0/1	No corresponde	No corresponde	
	G0/0/1.100	192.168.1.65	255.255.255.192	
	G0/0/1.200	192.168.1.97	255.255.255.224	
	G0/0/1.1000	No corresponde	No corresponde	
S1	VLAN 200	192.168.1.66	255.255.255.192	192.168.1.65
S2	VLAN 1	192.168.1.98	255.255.255.224	192.168.1.97
PC-A	NIC	DHCP	DHCP	DHCP
PC-B	NIC	DHCP	DHCP	DHCP

Paso 8: Asignar las VLAN a las interfaces del switch correctas
¿Por qué la interfaz F0/5 aparece en la VLAN 1?
Al ser la VLAN 1 la VLAN predeterminada, la interfaz fa0/5 se quedará ahí hasta ser configurada como enlace troncal o se asigne a otra VLAN.

Paso 9: Configure manualmente la interfaz F0/5 de S1 como un troncal 802.1Q.
En este punto, ¿qué dirección IP tendrían los PC si estuvieran conectados a la red usando DHCP? no tendrían ninguna ip válida, ya que el router no tiene configurado aún un servidor DHCP, eso se hace hasta la parte 2.

Switch S2
reinicio:

```
Switch#erase sta
Switch#erase startup-config
*****************************************************************
*****************************
Erasing Nvram will not clear license trust code.
*****************************************************************
*****************************
Erasing the nvram filesystem will remove all configuration fi
firm]
[OK]
Erase of nvram: complete
Switch#
```

```
S2(config-if)#interface vlan 1
S2(config-if)#ip address 192.168.1.98 255.255.255.240
S2(config-if)#exit
S2(config)#ip default-gateway 192.168.1.97
S2(config)#
S2(config)#
```

Router R1

reinicio:

```
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down

GigabitEthernet0/0         192.168.1.97    YES manual up                    up

GigabitEthernet0/1         unassigned      YES unset  up                    up

GigabitEthernet0/1.100     192.168.1.1     YES manual up                    up

GigabitEthernet0/1.200     192.168.1.65    YES manual up                    up

GigabitEthernet0/1.1000    unassigned      YES unset  up                    up
```

```
Router>
Router>enable
Router#erase star
Router#erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [con
firm]
[OK]
Erase of nvram: complete
Router#
*Aug 31 20:47:19.163: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#reload
Proceed with reload? [confirm]

*Aug 31 20:47:26.515: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
 Reload Command.
System Bootstrap, Version 15.0(1r)M16, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2012 by cisco Systems, Inc.

Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled


Readonly ROMMON initialized
program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340
```

configuración:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable secret class
R1(config)#line con 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#line vty 0 4
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#service password-encryption
R1(config)#banner motd #No entres#
R1(config)#exit
```

```
R1(config)#interface G0/0/0
                         ^
% Invalid input detected at '^' marker.

R1(config)#ip address 192.168.1.97 255.255.255.240
                    ^
% Invalid input detected at '^' marker.

R1(config)#no shutdown
% Incomplete command.

R1(config)#exit
R1#
*Aug 31 20:52:08.391: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#show ip int
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         unassigned      YES unset  administratively down down
GigabitEthernet0/1         unassigned      YES unset  administratively down down
R1#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface G0/0
R1(config-if)#ip address 192.168.1.97 255.255.255.240
R1(config-if)#no shutdown
R1(config-if)#
```

```
R1(config)#
R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
R1(config)#ip dhcp excluded-address 192.168.1.97 192.168.1.101
R1(config)#
R1(config)#ip dhcp pool Subred_A
R1(dhcp-config)#network 192.168.1.0 255.255.255.192
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#domain-name ccna-lab.com
R1(dhcp-config)#lease 2 12 30
R1(dhcp-config)#
R1(dhcp-config)#ip dhcp pool R2_Client_LAN
R1(dhcp-config)#network 192.168.1.96 255.255.255.240
R1(dhcp-config)#default-router 192.168.1.97
R1(dhcp-config)#domain-name ccna-lab.com
R1(dhcp-config)#lease 2 12 30
```

```
R1(config-if)#interface G0/1
R1(config-if)#no shutdown
R1(config-if)#interface G0/1.100
R1(config-subif)#description VLAN Cliente
R1(config-subif)#encapsulation dot1Q 100
R1(config-subif)#ip address 192.168.1.1 255.255.255.192
R1(config-subif)#interface G0/1.200
R1(config-subif)#description VLAN Administracion
R1(config-subif)#encapsulation dot1Q 200
R1(config-subif)#ip address 192.168.1.65 255.255.255.224
R1(config-subif)#interface G0/1.1000
R1(config-subif)#description VLAN Nativa
R1(config-subif)#encapsulation dot1Q 1000 native
R1(config-subif)#exit
R1(config)#
```

Configuración S1

Reinicio:

```


ROM: IOS-XE ROMMON
BOOTLDR: CAT3K_CAA Boot Loader (CAT3K_CAA-HBOOT-M) Version 6.08, RELEASE SOFTWAR
E (P)

S1 uptime is 1 hour, 39 minutes
Uptime for this control processor is 1 hour, 41 minutes
System returned to ROM by Reload Command
System restarted at 11:55:51 UTC Mon Aug 31 2026
System image file is "flash:packages.conf"
Last reload reason: Reload Command
```

```
Password:
S1>enable
Password:
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface vlan 200
S1(config-if)#ip address 192.168.1.66 255.255.255.224
S1(config-if)#exit
S1(config)#ip default-gateway 192.168.1.65
S1(config)#end
S1#
Aug 31 13:19:00.385: %SYS-5-CONFIG_I: Configured from console by console
S1#
Aug 31 13:25:32.962: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/6, changed state to down
Aug 31 13:25:33.966: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/6, changed state to down
Aug 31 13:25:37.190: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/6, changed state to up
Aug 31 13:25:38.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/6, changed state to up
Aug 31 13:27:03.860: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/6, changed state to down
Aug 31 13:27:04.863: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/6, changed state to down
Aug 31 13:27:12.707: %ILPOWER-5-DETECT: Interface Gi1/0/6: Power Device detected: IEEE PD
Aug 31 13:27:12.987: %ILPOWER-5-POWER_GRANTED: Interface Gi1/0/6: Power granted
Aug 31 13:27:13.697: %ILPOWER-5-PD_ENTRY_REMOVAL: Interface GigabitEthernet1/0/6: power device entry removed. admin_state: 2, oper_state: 2
Aug 31 13:27:13.702: %ILPOWER-5-IEEE_DISCONNECT: Interface Gi1/0/6: PD removed
Aug 31 13:27:13.705: %ILPOWER-5-IEEE_DISCONNECT: Interface Gi1/0/6: PD removed
Aug 31 13:27:19.530: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/6, changed state to up
Aug 31 13:27:20.530: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/6, changed state to up
Aug 31 13:27:26.702: %ILPOWER-5-DETECT: Interface Gi1/0/6: Power Device detected: IEEE PD
Aug 31 13:27:27.082: %ILPOWER-5-PD_ENTRY_REMOVAL: Interface GigabitEthernet1/0/6: power device entry removed. admin_state: 2, oper_state: 2
Aug 31 13:27:27.092: %ILPOWER-5-IEEE_DISCONNECT: Interface Gi1/0/6: PD removed
S1#
```

COM4 - PuTTY

```
Aug 31 12:35:10.201: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthe
rnet1/0/5, changed state to up
Aug 31 12:35:42.202: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan200, ch
anged state to up
S1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  unassigned      YES unset  up                    down
Vlan200                192.168.1.66    YES manual up                    up
GigabitEthernet0/0     unassigned      YES unset  down                  down
GigabitEthernet1/0/1   unassigned      YES unset  administratively down down
GigabitEthernet1/0/2   unassigned      YES unset  administratively down down
GigabitEthernet1/0/3   unassigned      YES unset  administratively down down
GigabitEthernet1/0/4   unassigned      YES unset  administratively down down
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  administratively down down
GigabitEthernet1/0/8   unassigned      YES unset  administratively down down
GigabitEthernet1/0/9   unassigned      YES unset  administratively down down
GigabitEthernet1/0/10  unassigned      YES unset  administratively down down
GigabitEthernet1/0/11  unassigned      YES unset  administratively down down
GigabitEthernet1/0/12  unassigned      YES unset  administratively down down
GigabitEthernet1/0/13  unassigned      YES unset  administratively down down
GigabitEthernet1/0/14  unassigned      YES unset  administratively down down
GigabitEthernet1/0/15  unassigned      YES unset  administratively down down
```

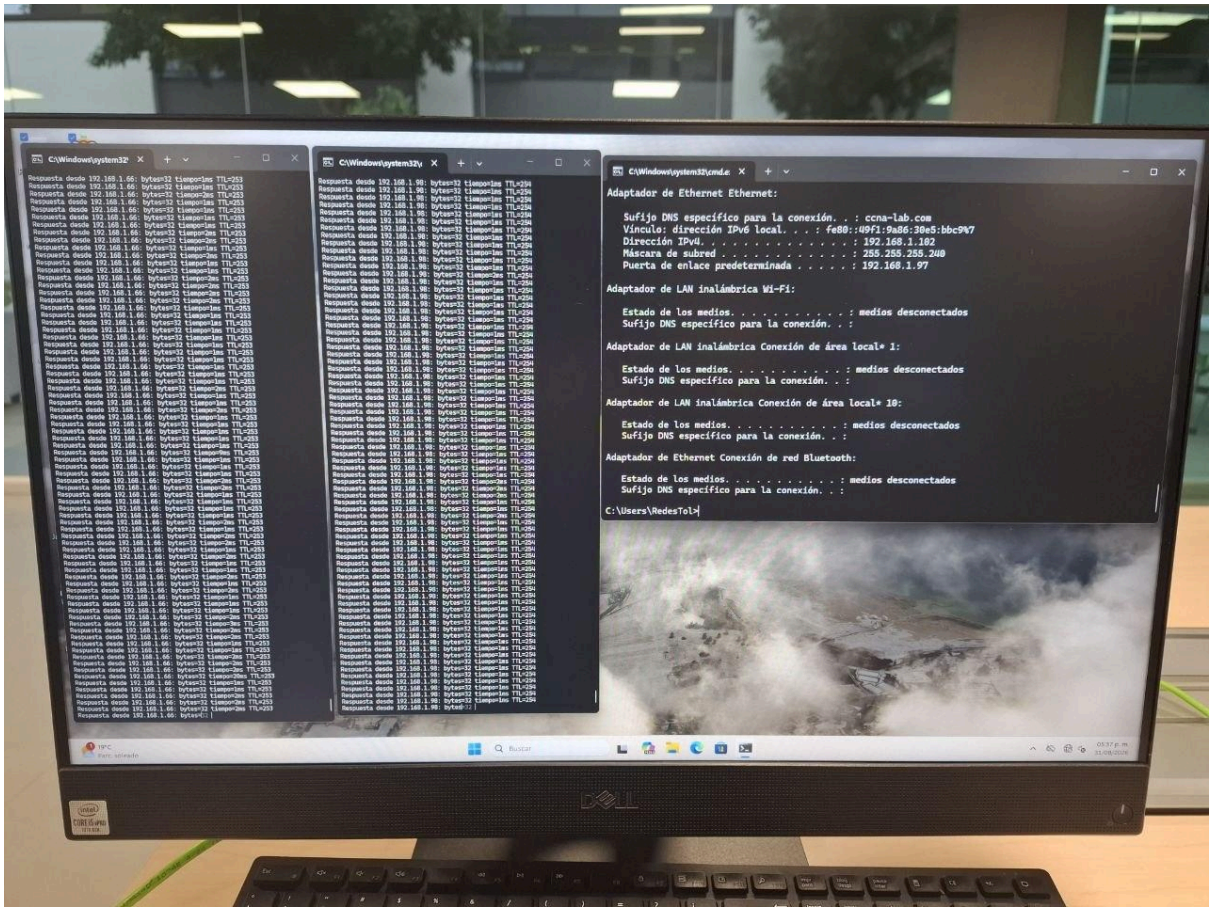
ping pc-A:

```
C:\Users\y4r1k>ping 192.168.1.102

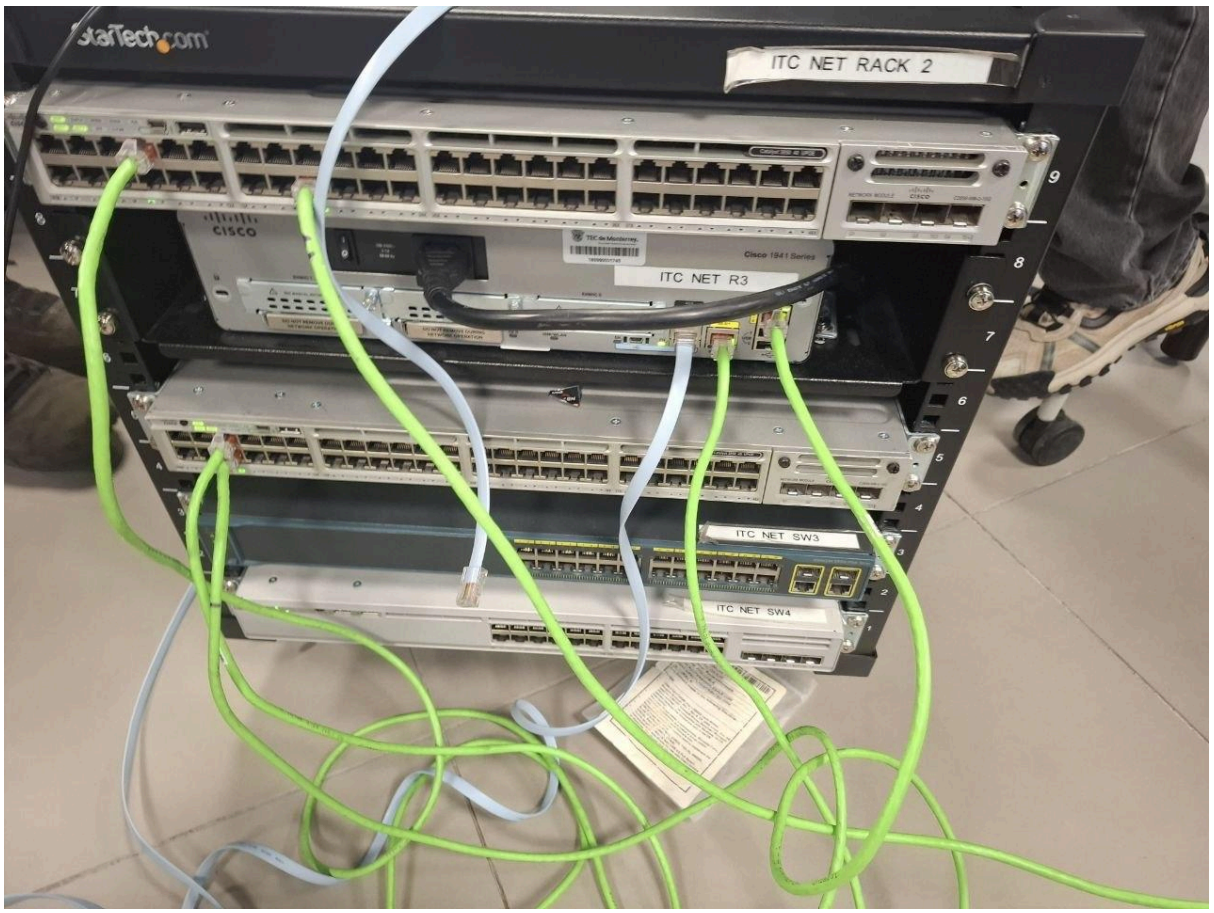
Haciendo ping a 192.168.1.102 con 32 bytes de datos:
Respuesta desde 192.168.1.102: bytes=32 tiempo=1ms TTL=127
Respuesta desde 192.168.1.102: bytes=32 tiempo=2ms TTL=127
Respuesta desde 192.168.1.102: bytes=32 tiempo=2ms TTL=127
Respuesta desde 192.168.1.102: bytes=32 tiempo=1ms TTL=127

Estadísticas de ping para 192.168.1.102:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 1ms, Máximo = 2ms, Media = 1ms
```

ping pc-B:


Adaptador de Ethernet Ethernet:
Sufijo DNS específico para la conexión. . : ccna-lab.com
Vínculo: dirección IPv6 local. . . : fe80::49f1:9a86:30e5:bbc9%7
Dirección IPv4. . . . . : 192.168.1.102
Máscara de subred . . . . . : 255.255.255.240
Puerta de enlace predeterminada . . . . . : 192.168.1.97
Adaptador de LAN inalámbrica Wi-Fi:
Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :
Adaptador de LAN inalámbrica Conexión de área local* 1:
Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :
Adaptador de LAN inalámbrica Conexión de área local* 10:
Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :
Adaptador de Ethernet Conexión de red Bluetooth:
Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :
C:\Users\RedesTol>

Cables:


ITC NET RACK 2
ITC NET R3
ITC NET SW3
ITC NET SW4